KERAAL Rehabilitation Research

1. Insights from the Research Proposal Especially the 2nd Paper

Question 1. What did previous KERAAL-related studies investigate?

The **KERAAL project** focuses on using human skeleton and movement data to analyze physical rehabilitation exercises for people with low-back pain. The KERAAL dataset contains rehabilitation recordings, skeleton information, videos, and medical annotations related to exercise correctness and errors.

The main studies and datasets related to our project are:

1. **KERAAL Dataset Study**
   Provides the main rehabilitation dataset, including skeleton sequences, videos, and expert annotations.
2. **Skeleton-Based Transformer Study**
   Uses a Transformer-based model to classify different rehabilitation exercise errors using skeleton data from KERAAL.
3. **Rotation-Invariant Skeleton Research**
   Investigates skeleton representations that are less affected by changes in body position or orientation.
4. **KIMORE Dataset**
   Provides rehabilitation exercise data, including RGB, depth, and skeleton joint information for low-back-pain exercises.
5. **UI-PRMD Dataset**
   Provides skeletal movement and joint-angle data for physical rehabilitation exercises and is useful for computational assessment of rehabilitation movements.

Question 2. What are the important findings from the second paper?

The second paper is:

Marusic, A., Nguyen, S. M., & Tapus, A. (2025). ***Skeleton-Based Transformer for Classification of Errors and Better Feedback in Low Back Pain Physical Rehabilitation Exercises.***

The study proposes a **Transformer-based model** for classifying rehabilitation exercise errors using skeleton-based movement information. The model was evaluated using the KERAAL dataset.

Important findings for our project are:

- The model can classify different types of rehabilitation errors.
- It uses **Kinect V2 skeleton data**.
- It uses a **Transformer-based model** for error classification.
- The study investigates the **importance of different joints** for recognizing exercise errors.
- The evaluation includes a **subject-independent setting**, which is important when testing whether a model can generalize to people it has not seen during training.
- The dataset contains **imbalanced error classes**, so accuracy alone may not fully describe model performance.
- The difference between **random splitting** and **subject-independent splitting** is important.
- There is also a **simulated-to-real problem**: healthy participants may deliberately perform errors, while patients may naturally produce errors during rehabilitation.

The paper therefore gives us a strong starting point for investigating whether better movement representations can improve rehabilitation-error classification.

Sources for the second paper:

- Full paper — arXiv
- Published paper — PubMed
- Official KERAAL code and dataset repository — GitHub

Question 3. What research opportunity comes from these findings?

Our project can investigate whether **kinematic features** such as:

- Joint angles
- Velocity
- Symmetry
- Other movement-related features

can improve rehabilitation-error classification compared with using only raw skeleton coordinates.

We can also investigate whether the important joints or features identified by the model correspond to the **body region associated with the annotated error**.

Therefore, our main research direction is:

> **Under subject-independent and simulated-to-real evaluation, do kinematic features aligned with error definitions improve rehabilitation-error classification?**

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2. Review of the KERAAL Dataset

Question 1. What are the error categories?

The KERAAL dataset contains three rehabilitation exercises, with three main error types for each exercise.

Exercise	Error 1	Error 2	Error 3
Torso Rotation (RTK)	Arms not raised enough	Torso rotation not sufficient	Body leaned to the side
Flank Stretch (ELK)	Opposite arm not along the body	Body not tilted	Upper arm not bent
Hiding Face (CTK)	Arms not raised enough	Arms not outspread enough	Description needs verification

Important: The wording of Error 1 and Error 3 for the **Hiding Face (CTK)** exercise should be checked against the original KERAAL labeling guide before final submission.

Question 2. What are the body-region categories?

The annotation scheme contains four main body-region categories:

- **LeftArm**
- **RightArm**
- **BothArms**
- **Spine**

The KERAAL documentation describes expert annotations associated with exercise correctness, error recognition, and the spatial and temporal localization of errors.

Question 3. Does every error have a body-region label?

No.

This is an important finding for our research.

Groups **1A and 2A** contain body-region labels, while **Group 3**, which we plan to use for training, contains error labels without body-region labels.

Therefore, **body-region prediction should initially be treated as an exploratory task rather than the main task**.

Question 4. Can we connect each label to a specific skeleton sequence?

Yes, in principle.

The KERAAL files use recording identifiers to connect the different data types and annotations. The dataset repository provides documentation and code for working with the KERAAL recordings and skeleton data.

This allows us to connect:

Participant → Exercise → Recording → Skeleton Sequence → Error Annotation

This connection is necessary before training a machine-learning model.

The KERAAL repository describes the dataset as containing recordings from healthy participants and patients with low-back pain, together with expert annotations and multiple forms of skeleton/movement data.

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3. Designing Our Study Based on Existing Research

Question 1. What five studies/datasets are relevant to our research?

Study 1 — KERAAL

KERAAL provides the main data for our project. It contains low-back-pain rehabilitation exercises, skeleton sequences, videos, and medical annotations.

Source:
KERAAL Dataset and GitHub Repository

Study 2 — Skeleton-Based Transformer

Marusic et al. developed a Transformer-based approach for classifying rehabilitation errors using skeleton data from KERAAL. This is especially important because our research can directly build upon this work.

Sources:
Skeleton-Based Transformer — arXiv
Skeleton-Based Transformer — PubMed

Study 3 — Rotation-Invariant Skeleton Methods

Related research on rotation-invariant skeleton representations motivates us to investigate better skeleton representations instead of relying only on raw coordinates.

For our project, this supports investigating **normalized and kinematic representations** that may be less affected by differences in body position or orientation.

Note: The exact bibliographic citation should be taken from the original project reference list before submitting the homework.

Study 4 — KIMORE

KIMORE is a rehabilitation movement dataset containing RGB, depth, and skeleton joint information. It includes five physical exercises specifically selected for low-back-pain rehabilitation and provides clinically relevant movement features.

Source:
KIMORE Dataset — IEEE

Study 5 — UI-PRMD

UI-PRMD is a physical rehabilitation movement dataset containing skeletal positions and joint angles from rehabilitation exercises. It provides useful data for studying automated assessment of rehabilitation movements.

Source:
UI-PRMD Dataset Paper — PMC

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4. What Is Our Main Research Question?

Our main research question is:

> **Do normalized kinematic features improve subject-independent rehabilitation-error classification compared with raw skeleton coordinates?**

We can compare the following representations:

Raw coordinates
↓
Joint angles
↓
Joint angles + velocity
↓
Joint angles + velocity + symmetry

This will help us determine whether adding meaningful movement information improves error classification.

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5. What Experiments Should We Perform?

Experiment 1 — Raw Skeleton Coordinates

Skeleton coordinates → Model → Error classification

This provides our **baseline**.

The baseline tells us how well the model performs before adding kinematic features.

Experiment 2 — Kinematic Features

Use features such as:

- Joint angles
- Velocity
- Symmetry
- Other movement-related features

Then test:

Kinematic features → Model → Error classification

We can compare the results with Experiment 1.

Experiment 3 — Body-Region Prediction

Test whether the model can identify the body region associated with an error.

Kinematic features → Model → Body-region prediction

This is useful for investigating whether the model focuses on the same body region identified by the medical annotations.

Experiment 4 — Multi-Task Learning

One model predicts both:

Error type + Body region

However, this experiment is exploratory because Group 3 does not contain body-region labels.

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6. How Should We Evaluate the Study?

Our main evaluation protocol is:

Training: Group 3
Testing: Groups 1A + 2A

This follows the general subject-independent setting used in the previous research and makes our results easier to compare with the previous study.

We should report:

- **Macro-F1**
- **Balanced Accuracy**
- **Per-class Recall**
- **Confusion Matrix**
- **Accuracy**
- **Majority-class baseline**

We should also compare:

Random Split vs. Subject-Disjoint Split

A random split can place recordings from the **same participant** into both the training and testing sets.

A subject-disjoint split prevents this and provides a stronger test of whether the model can generalize to **unseen participants**.

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7. Final Research Structure

Our final research paper can be organized as follows:

1. Introduction

Explain why automatic rehabilitation-error detection is useful.

2. Related Work

Discuss:

- KERAAL
- Skeleton-Based Transformer
- Rotation-invariant skeleton methods
- KIMORE
- UI-PRMD

3. Dataset

Explain:

- KERAAL groups
- Exercises
- Error labels
- Body-region labels
- Skeleton data

4. Method

Explain:

- Raw skeleton coordinates
- Joint angles
- Velocity
- Symmetry
- Feature normalization

5. Experiments

Compare:

- Raw coordinates
- Angles
- Angles + velocity
- Angles + velocity + symmetry
- LSTM / TCN models
- Body-region analysis

6. Results

Report:

- Macro-F1
- Balanced accuracy
- Recall
- Accuracy

- Confusion matrices

7. Discussion

Discuss:

- Which features improve performance
- Which errors are easier or harder to classify
- Whether important model features correspond to annotated body regions
- Whether the model generalizes to unseen participants

8. Limitations

Discuss limitations such as:

- Small training group
- Class imbalance
- Missing body-region labels in Group 3
- Difference between healthy simulated errors and real patient errors

9. Conclusion

Summarize whether kinematic features provide useful information for rehabilitation-error classification.

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8. Main Sources

KERAAL

KERAAL Dataset — Official GitHub Repository

The repository describes KERAAL as a medical database for low-back-pain rehabilitation, including skeleton sequences, videos, and expert annotations.

Skeleton-Based Transformer

Skeleton-Based Transformer — arXiv

Skeleton-Based Transformer — PubMed

KIMORE

KIMORE Dataset — IEEE

UI-PRMD

UI-PRMD Dataset Paper — PubMed Central